LEGALVISION: A KNOWLEDGE-DRIVEN AI FRAMEWORK FOR LEGAL UNDERSTANDING AND TRUST ASSESSMENT

25-26J-127

Project Proposal Report

WICRAMASINGHE D A T N

B.Sc. (Hons) Degree in Information Technology Specialized in Data Science

Department of Information Technology

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DECLARATION

I declare that this is my own work and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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The supervisor/s should certify the proposal report with the following declaration.

The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

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Signature of the supervisor Date
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....2025/8/28
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ABSTRACT

Understanding and interpreting lengthy legal documents is a major challenge, particularly for non-experts who often find legal language too complex and inaccessible. Even legal professionals face difficulties in extracting information efficiently due to the volume and density of legal text. This study addresses this problem by proposing a system that provides **multi-perspective summarization and visualization of legal documents**, designed to enhance accessibility, transparency, and comprehension. The overall purpose of the research is to bridge the gap between legal expertise and public understanding by offering an intelligent and user-friendly solution.

The proposed system is structured around three main steps. First, users can upload legal documents in standard digital formats. Second, the system applies advanced natural language processing techniques to generate **multi-perspective summaries**, which include case study-focused summaries, financial or practical summaries, procedural guidance summaries, and rights and duties-based summary. Third, the system enhances these textual outputs with **interactive visualization features**. To ensure transparency and trust, users can link back from any summarized section to the **original clause** in the uploaded document. Beyond summaries, the system also generates **infographics for important legal terms**, helping users quickly grasp the meaning of critical concepts. Additionally, it provides **interactive charts**, such as risk distribution graphs and timeline views, to present trends and legal events in an intuitive format.

Legal relationships and clause dependencies are further illustrated through **graph-based diagrams using Graphviz DOT language**, enabling both narrative and structural exploration of the document. By combining multi-perspective summarization with visual outputs, the proposed approach not only reduces cognitive overload but also supports traceability, transparency, and practical decision-making. Ultimately, this system contributes to making legal information more **accessible, interpretable, and actionable** for both legal experts and non-expert stakeholders alike.

Keywords: Legal documents, Multi-perspective summarization, Visualization, Transparency, Infographics, Risk distribution, Timeline analysis, Graphviz, Natural Language Processing

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LIST OF ABBREVIATIONS

Abbreviation	Description
AI	AI Artificial Intelligence
API	API Application Programming Interface
BERT	Bidirectional Encoder Representations from Transformers
GPU	GPU Graphics Processing Unit
NER	Named Entity Recognition
NLP	NLP Natural Language Processing
RAG	RAG Retrieval-Augmented Generation
ROUGE	Recall-Oriented Understudy for Gisting Evaluation
SSD	SSD Solid State Drive
T5	Text-to-Text Transfer Transformer

1. INTRODUCTION

1.1. Background & Literature survey

1.1.1. Background

The proposed research develops an integrated, user-centered AI system for property law, aiming to bridge the gap between complex legal documents and public understanding. Existing legal AI systems face critical limitations: they provide only single-perspective summaries [1], lack traceability mechanisms [2], and often produce inaccurate visualizations [3] without contextual awareness of legal relationships. To address these gaps, the **proposed system integrates modules for legal document ingestion, multi-perspective summarization, risk and bias detection, knowledge graph construction, and visualization.** Collectively, these modules generate simplified yet trustworthy interpretations of property law documents, enabling non-expert users such as property owners, buyers, and citizens to better understand their rights, responsibilities, and risks. By combining summarization, explainability, and visual analytics, the system promotes transparency, accessibility, and usability of property law for everyday decision-making.

Within this broader system, my component focuses on **multi-perspective summarization and visualization**. It generates summaries from diverse perspectives, including case-focused, financial, procedural, and responsibility views, while ensuring traceability by linking each summary back to the original legal clause. Beyond text outputs, the component incorporates visualization features such as risk distribution charts, infographics, timelines, and graph-based clause representations using Graphviz. This dual text-visual approach helps non-experts move beyond surface-level summaries and engage with property law in intuitive, multi-format ways.

The specific aim of my component is to provide a comprehensive yet accessible tool that (1) generates clear, multi-perspective summaries, (2) ensures transparency and trust via traceability, and (3) enhances understanding through visual explanations of clauses, risks, and relationships.

1.1.2. Relevant Studies and Techniques

Multi-perspective summarization of legal documents has emerged as a key research area, with several foundational studies shaping theoretical and practical frameworks. The survey *“A Comprehensive Survey on Legal Summarization: Challenges and Future Directions”* (2025) [1] analyzed over 120 papers from the transformer era of NLP, **highlighting state-of-the-art techniques and identifying challenges in generating perspective-aware summaries**. This builds upon Kanapala et al.’s foundational survey *“Summarization of legal documents: Where are we now and the way forward”* [4], which **categorized approaches into extractive vs. abstractive and supervised vs. unsupervised methods**, providing theoretical groundwork for generating case-focused, financial, procedural, and responsibility-based perspectives.

Visualization in legal document analysis has advanced through works like *“Enhancing the Visualization of Law”* [5], which introduced **methods for improving readability** via semantic information extraction and model enrichment. Further, *“LegalViz: Legal Text Visualization by Text to Diagram Generation”* [3] presented **techniques for creating diagrams** from complex corpora, identifying entities, rules, statements, and transactions graph-based clause representations using Graphviz.

The **methodological challenges** were addressed in *“Legal Case Document Summarization: Extractive and Abstractive Methods and their Evaluation”* [6], which addresses the challenging problem of maintaining semantic coherence while extracting different viewpoints from legal judgments.

These techniques should be mastered through a combination of studying foundational surveys and state-of-the-art transformer models**,** implementing hands-on experiments with extractive, abstractive, and perspective-aware summarization methods**,** and practicing semantic extraction with knowledge graphs and visualization tools like Graphviz**.** Additionally, applying evaluation frameworks on real legal corpora will help refine both summarization quality and visualization effectiveness.

1.1.3. State of the Art

Current state-of-the-art legal document summarization employs hybrid transformer architectures, with 2025 developments **using double-model approaches** that merge T5 for abstractive summarization with BERT for extractive summarization, achieving ROUGE-1 scores of 0.538 [1]. Advanced frameworks integrate dynamic Retrieval-Augmented Generation (**RAG**) with domain-specific adaptation, leveraging BM25 retrievers and top-3 chunk selection for accurate legal knowledge extraction [7]. In visualization, LegalViz [3] **benchmarks datasets** of 23 languages and 5,580 document-visualization pairs using Graphviz DOT, producing diagrams identifying legal entities, rules, and transactions. Innovations like Deep Clustering Enhanced Summarization (DCESumm) [6] handle long documents, while knowledge graph-based legal document generation [8] surpasses rule-based methods, enabling sophisticated **graph-based representations**. These advances bridge complex legal content with user comprehension, supporting nuanced multi-perspective understanding in specialized domains.

1.1.4. How Others Have Tried to Solve These Problems and How our Approach Differs

Previous research efforts in legal document analysis have addressed individual components but failed to provide comprehensive solutions. Traditional approaches to legal summarization have relied on single-reference summary datasets [1] and generic summarization models, producing uniform outputs that **lack the multi-perspective analysis required for property law comprehension**. Existing systems like those documented in recent surveys generate summaries focused on case precedents or procedural aspects in isolation, without considering financial implications or responsibility distributions that property law require.

Visualization efforts in legal AI have predominantly focused on basic network representations and entity-relationship diagrams, but existing systems **generate inaccurate** [3] **or oversimplified structural charts** without proper traceability to source clauses or contextual understanding of legal relationships. Current legal document visualization tools produce **generic graphics** [9] that display legal connections but lack the precision and verification mechanisms needed to ensure reliability, often failing to integrate comprehensive textual analysis with meaningful visual representations that accurately reflect complex property law relationships.

Multi-perspective summarization and visualization component addresses these limitations through integrated innovation. Unlike previous work that treats summarization and visualization as **separate processes** [1], therefore approach combines perspective-aware content generation with accurate, traceable visual representation, enabling simultaneous production of case-focused, financial, procedural, and responsibility summaries linked to precise entity-relationship diagrams and enhanced visualizations. The **traceability mechanism addresses** critical gaps where existing systems cannot verify generated insights against source clauses [2], while our enhanced visual representations using risk distribution charts, interactive timelines, and Graphviz-based graphs provide accurate, contextually enriched diagrams with complete traceability rather than displaying unreliable structural relationships without verification capabilities.

1.2. Research Gap

Existing legal summarization systems face several limitations that hinder accessibility, interpretability, and comprehensiveness. First, there is a lack of multi-reference summary datasets [1], restricting the development and evaluation of robust summarization methods. Second, **there is no support for multi-perspective summarization** [1], preventing generation of diverse outputs such as case-focused, financial, procedural, and rights-and-duties summaries. Third, legal jargon and inconsistent terminology remain barriers for non-experts [1] - [2], as systems rarely preprocess terms into plain, consistent language, reducing accessibility. Fourth, query-based methods are limited [10], often retrieving sentences without producing coherent, document-level summaries and failing to preserve legal context, particularly in property law. Fifth, **transparency and accountability remain underexplored** [2], with most systems lacking traceability between summaries and original clauses. Finally, **visual interpretation of legal text is a critical gap** [3], [9], as existing systems produce oversimplified or inaccurate graphs without enriched infographic-style visualizations that capture context, proportions, and clause relationships. These gaps underscore the need for a framework producing **multi-perspective, plain-language, and traceable summaries** with **accurate, reliable, and context-enriched visualizations**.

This research will develop a multi-perspective summarization framework that produces diverse summaries (case study, financial, procedural, rights and duties) supported by a legal-to-plain language pipeline for accessibility. It will generate coherent, document-level outputs that preserve property law context, ensure transparency through traceability to original clauses, and enhance visualization by creating accurate, context-aware diagrams and enriched infographics that clearly represent legal obligations, dependencies, and proportions for both experts and non-experts.

1.3. Research Problem

Legal documents are often complex, dense, and difficult for non-experts to interpret, which limits their accessibility and usability. Existing legal summarization systems face several challenges: they lack multi-perspective summaries, struggle with legal jargon, and generate visualizations that are neither fully accurate nor traceable to original clauses. This results in inefficiencies, misinterpretations, and reduced trust among users relying on legal summaries.

- **How can multi-perspective summarization and interactive visualization be integrated to make complex legal documents accessible to non-experts through traceable summaries, dynamic charts, infographics, and graph-based diagrams?**

2. OBJECTIVES

2.1. Main Objectives

To develop a legal document summarization and visualization system that generates accurate, multi-perspective summaries and converts textual information into reliable, traceable, and informative visual representations, thereby improving accessibility and interpretability of legal documents for non-experts.

2.2. Specific Objectives

1. **To design a preprocessing and summarization framework** that simplifies complex legal jargon into standardized language and generates multi-perspective summaries (case study, financial/practical, procedural, and rights-and-duties-based).
2. **To ensure transparency and interpretability** by integrating traceability mechanisms that link summaries back to original clauses and by adopting multimodal approaches combining textual and visual representations.
3. **To enhance comprehension through visualization** by generating diagrams, enriched infographics, and dynamic charts/timelines that illustrate key clauses, risks, and temporal relationships in an accessible manner.
4. **To evaluate and refine the system's effectiveness** in terms of accuracy, interpretability, and user-friendliness, particularly for non-expert audiences, while extracting actionable insights from legal documents.

3. METHODOLOGY

3.1. Problem Identification and Requirements Analysis

This phase begins with comprehensive stakeholder analysis through **surveys among Sri Lankan general public to identify priority legal areas**, focusing on property law accessibility challenges. Structured interviews with legal professionals, property owners, and end-users reveal current pain points in legal document comprehension. **Requirements analysis** defines functional specifications for multi-perspective summarization (case-focused, financial, procedural, rights-based summaries), visualization capabilities (risk distribution charts, timelines, infographics with interactive features), and traceability mechanisms using unique identifiers linking summaries to original clauses. Performance metrics establish accuracy thresholds (targets 85% summarization quality), response times (30-second generation), and user satisfaction targets (80% helpfulness rating), while specifying data collection requirements including document formats (PDF, DOCX, TXT) and legal terminology databases

3.2. System Design and Architecture Development

The system design phase creates a **modular architecture** integrating multiple components for simplifying property law documents through component-based approach with detailed system flow and data flow models. The system architecture is illustrated Figure 3.1:System Diagram. The **preprocessing module** handles text cleaning, entity recognition, and legal clause identification using advanced NLP techniques. The **summarization engine** incorporates algorithms generating case study-focused, financial, procedural, and rights and duties-based summaries. The **visualization engine** emphasizes user-friendly diagrams using Graphviz DOT language, complemented by interactive charts and timelines including clause distribution charts, case flow timelines, risk distribution charts, and infographics of important legal terms. A **transparency module** ensures bidirectional linking between summaries and original clauses. The web-based interface prioritizes responsive design and accessibility standards.

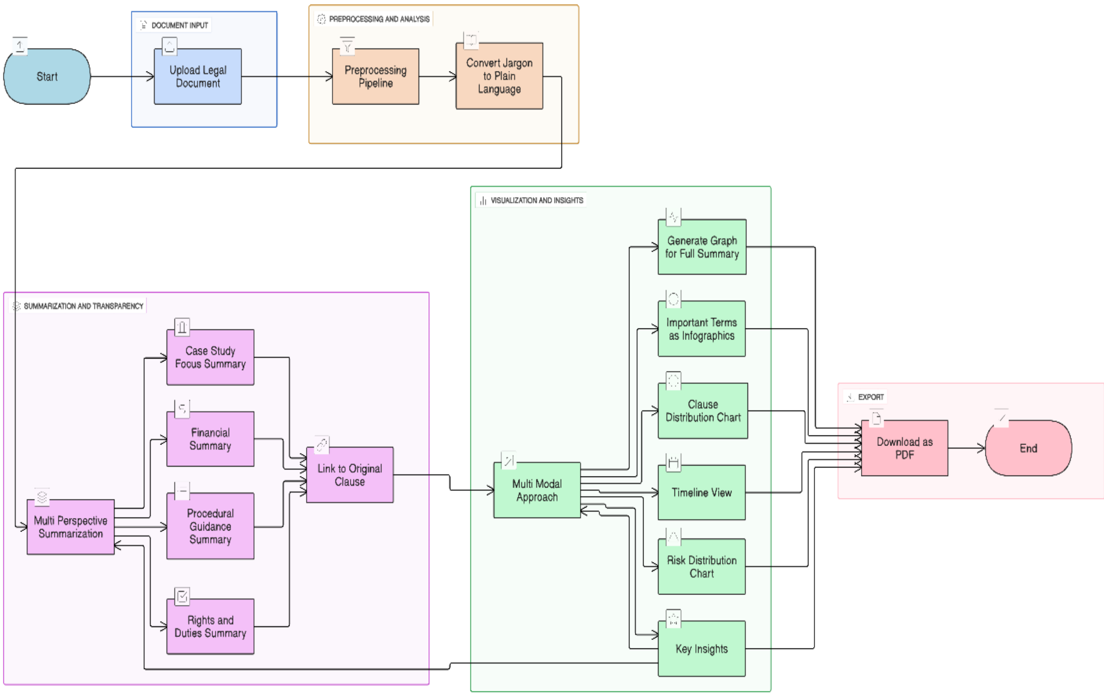

Figure 3.1:System Diagram

3.3. Prototype Development and Implementation

The prototype development follows **Agile methodology** with incremental implementation using **Python Flask** backend, **spaCy/NLTK** for NLP processing, and fine-tuned **BERT/LegalBERT** models for legal text processing. The document processing pipeline handles PDF/DOCX parsing, implements Named Entity Recognition for legal entities, and performs clause segmentation. Multi-perspective summarization uses transformer-based models generating case study summaries (key facts, outcomes), financial summaries (costs, obligations), and procedural summaries (required steps, timelines). Visualization components utilize Graphviz for document structure diagrams, **D3.js** and **Plotly** for interactive timelines, risk assessment charts, and financial breakdowns. Transparency features ensure accurate bidirectional linking between simplified content and original clauses with confidence scoring mechanisms. Additionally, the system integrates the **Canvas API** to automatically generate professional infographics, enabling visually appealing representations of important terms, and key contract highlights.

3.4. System Evaluation and Validation

Evaluation methodology employs both quantitative and qualitative assessment approaches. Performance evaluation includes accuracy assessment using **ROUGE** scores for summarization quality, precision-recall metrics for key insight extraction, and response time measurements for system efficiency. User experience validation involves conducting usability testing sessions with non-expert participants, measuring comprehension improvement through pre/post-test assessments, and gathering feedback on visualization effectiveness. Expert validation includes review sessions with legal professionals to ensure legal accuracy and compliance with Sri Lankan property law standards

3.5. Data Management and Ethical Considerations

Data management protocols ensure secure handling of legal documents and user information throughout the system lifecycle. Document collection procedures establish partnerships with **legal institutions in Sri Lanka** for accessing diverse property law documents while maintaining confidentiality agreements. Data collection for system involves gathering legal documents **from lawyers and law colleges in Sri Lanka**, ensuring diverse document types and complexity levels representative of real-world property law scenarios. Database design in MongoDB incorporates encryption for sensitive legal content and user privacy protection measures. Ethical considerations include informed consent procedures for survey participants, anonymization protocols for legal documents, and compliance with Sri Lankan data protection regulations.

3.6. System Refinement and Final Optimization

Final optimization phase focuses on performance enhancement and user experience refinement based on evaluation results. System performance tuning includes algorithm optimization for faster processing speeds, **database query optimization** for improved response times, and **frontend optimization** for better user interaction. Based on user feedback and expert reviews, interface improvements include enhanced visualization clarity, improved navigation flows, and additional customization options for different user types. **Integration testing** ensures seamless operation across all system components, while stress testing validates system performance under high-load conditions. Documentation development includes user manuals, technical specifications, and deployment guides for future maintenance and updates

4. REQUIREMENTS

4.1. Functional Requirements

1. **Document Processing**
   - The system shall accept legal documents in multiple formats, extract text, identify clauses, and convert legal jargon into plain language.
2. **Multi-Perspective Summarization**
   - The system shall generate case-focused, financial, procedural, and rights-and-duties-based summaries, and allow comparative analysis.
3. **Traceability and Transparency**
   - The system shall provide bidirectional linking between summary sentences and original clauses with confidence indicators and highlighting.
4. **Visualization Generation**
   - The system shall create interactive diagrams, charts, timelines, and infographics to represent legal information with navigation support.
5. **Key Insight Extraction**
   - The system shall automatically extract and highlight critical legal information and present key insights in structured formats.

4.2. Non-Functional Requirements

1. **Performance**
   - The system shall process large legal documents efficiently, generate summaries quickly, and render visualizations with minimal delay while supporting multiple users simultaneously.
2. **Accuracy**
   - The system shall ensure high-quality summarization, accurate terminology conversion, and reliable traceability between summaries and original clauses.

3. **Usability**
    - The system shall be user-friendly, accessible to non-legal users, require minimal training, and comply with accessibility standards.
4. **Reliability**
    - The system shall provide high availability, perform regular backups, and ensure quick recovery from failures.
5. **Security**
    - The system shall protect data using strong encryption, enforce secure user sessions, maintain audit logs, and undergo regular security assessments.

4.3. System Requirements

1. **Hardware Requirements**
    - The system shall require a minimum of 16 GB RAM, 8-core CPU, GPU support, and 500 GB SSD storage with horizontal scaling capability.
2. **Software Requirements**
    - The system shall use Python 3.8+ with Flask 2.0+, MongoDB 4.4+, React 18+ with Node.js 16+, spaCy 3.4+, and Graphviz 2.50+.
3. **System Architecture**
    - The system shall implement a microservices backend with Python Flask, a React.js frontend with D3.js/Chart.js for visualizations, a MongoDB database, and Docker-based containerized deployment.

4.4. User Requirements

1. **Ease of Use**
    - Users shall be able to upload legal documents in common formats (PDF, DOCX, TXT) without requiring technical knowledge.
2. **Accessibility**

- Users shall receive summaries in plain, easy-to-understand language, even if they are non-legal experts.

3. **Traceability**
    - Users shall be able to verify summaries by tracing each sentence back to its original clause with a single click.
4. **Interactivity**
    - Users shall be able to filter, zoom, and drill down into visualizations for deeper exploration of legal content.
5. **Reliability & Security**
    - Users shall be assured that their documents are processed securely and remain confidential during system use.

4.5. Test Cases

1. Document Upload Validation

   Input: Upload PDF property deed (25 pages)
   Expected Output: Document processed successfully, confirmation message displayed
   Test Type: Functional

2. Multi-Perspective Summary Generation

   Input: Processed property contract
   Action: Generate financial summary
   Expected Output: Summary highlighting costs, payment schedules, financial obligations with 85%+ accuracy
   Test Type: Functional

3. Traceability Link Accuracy

Input: Generated case-focused summary
Action: Click on summary sentence about property boundaries
Expected Output: Original clause highlighted, 99% mapping accuracy
Test Type: Functional

4. Visualization Rendering

Input: Risk distribution data from property analysis
Expected Output: Interactive chart rendered within 15 seconds, all interactive features functional
Test Type: Functional/Performance

5. Mobile Responsiveness

Input: Access system via mobile browser
Expected Output: Interface adapts to screen size, all features accessible, touch-friendly navigation
Test Type: Usability

5. COMMERCIALIZATION

The commercialization strategy for **LEGALVISION** focuses on transforming the research prototype into a scalable, market-ready solution that addresses real-world legal challenges. Given the increasing complexity of legal documentation and the demand for accessible legal services, there is strong potential for adoption across multiple target groups.

Target Market

- **Property Owners and Buyers** – to understand agreements, deeds, and contractual obligations.
- **Small Business Owners** – to interpret contracts, rental agreements, and service terms without relying on expensive legal consultations.
- **General Citizens** – to gain awareness of rights, responsibilities, and procedures in everyday legal documents.

Value Proposition

- **Comprehensive Understanding** – Delivers multiple perspectives (case, financial, procedural, rights & duties).
- **Risk Awareness** – Highlights risks, biases, and responsibilities within documents.
- **Trust & Transparency** – Ensures every summary is traceable to original clauses.
- **Simplified Visuals** – Provides infographics, knowledge graphs, and timelines for intuitive learning.
- **Accessibility** – Designed for non-experts with plain-language explanations and multilingual support.

Revenue Model

- **Freemium Platform** – Basic document upload and summarization available for free; advanced features (visualizations, risk/bias detection, downloads) offered under a low-cost premium plan.
- **Community Version** – A free awareness-focused version distributed via universities and community centers.

LEGALVISION is a citizen-focused system designed to make legal documents accessible to the general public, unlike existing tools built mainly for lawyers and enterprises. It combines plain-language translation, traceability of summaries to original clauses, risk and bias detection, and visualizations such as timelines, graphs, and infographics. The commercialization strategy begins with a pilot launch in Sri Lanka focusing on property law, followed by a web with summarization and visualization features. Public awareness campaigns and universities, and government institutions will support adoption. In the long term, LEGALVISION aims to expand into broader legal domains and international markets to improve legal accessibility worldwide.

6. OVERALL SYSTEM DIAGRAM

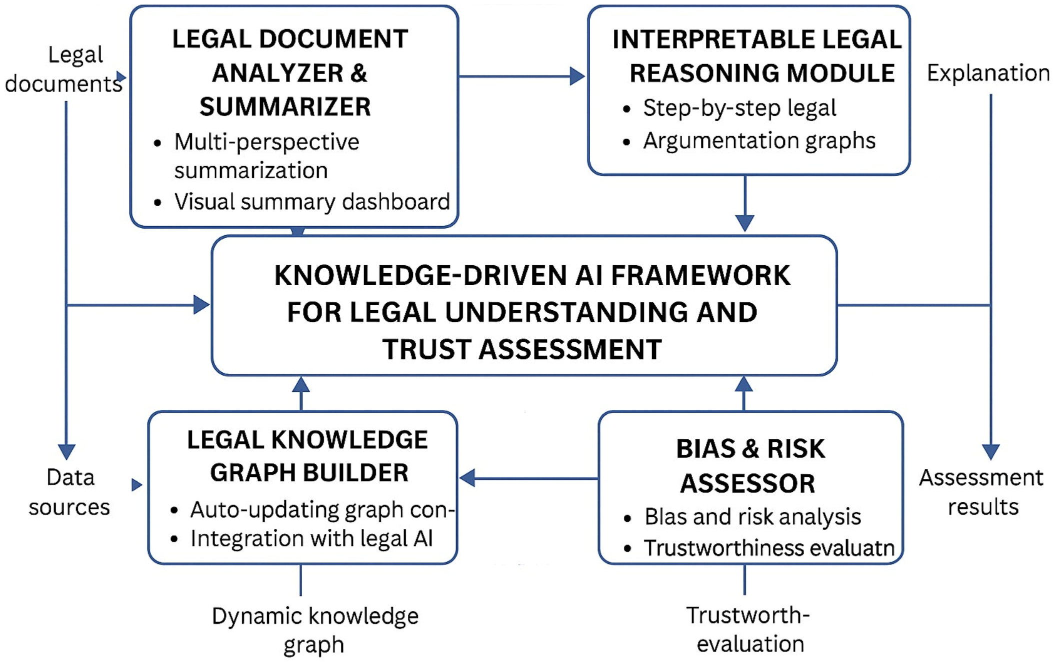

Figure 6.1: Overall System Diagram

Figure 6.1: Overall System Diagram shows the overall system architecture, where legal documents are analyzed and summarized into multiple perspectives. The framework ensures transparency through interpretable reasoning, builds dynamic knowledge graphs, and assesses bias and risks to provide trustworthy and comprehensive legal insights.

7. CONTEXT DIAGRAM

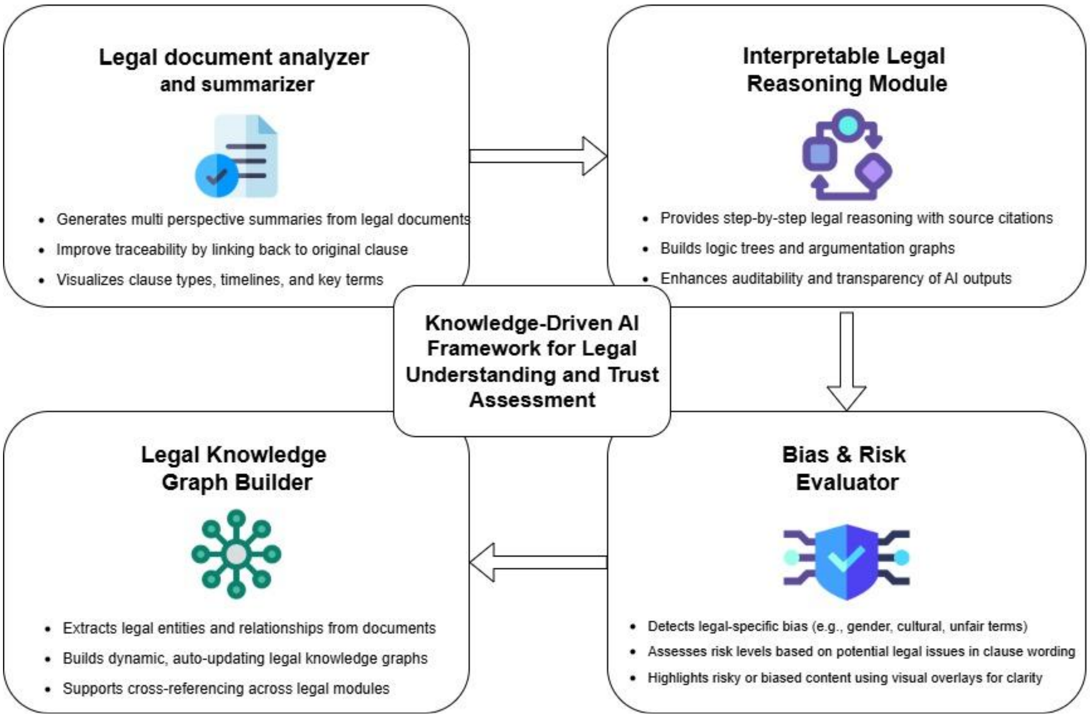

Figure 7.1: Context Diagram

Figure 7.1: Context Diagram presents the context diagram of the proposed framework. It highlights how the system processes legal documents through four core modules: summarization, interpretable reasoning, knowledge graph building, and bias/risk evaluation. These components work together to generate clear, transparent, and trustworthy legal insights.

8. GANTT CHART

Figure 8.1: Gantt Chart

9. WORK BREAKDOWN CHART

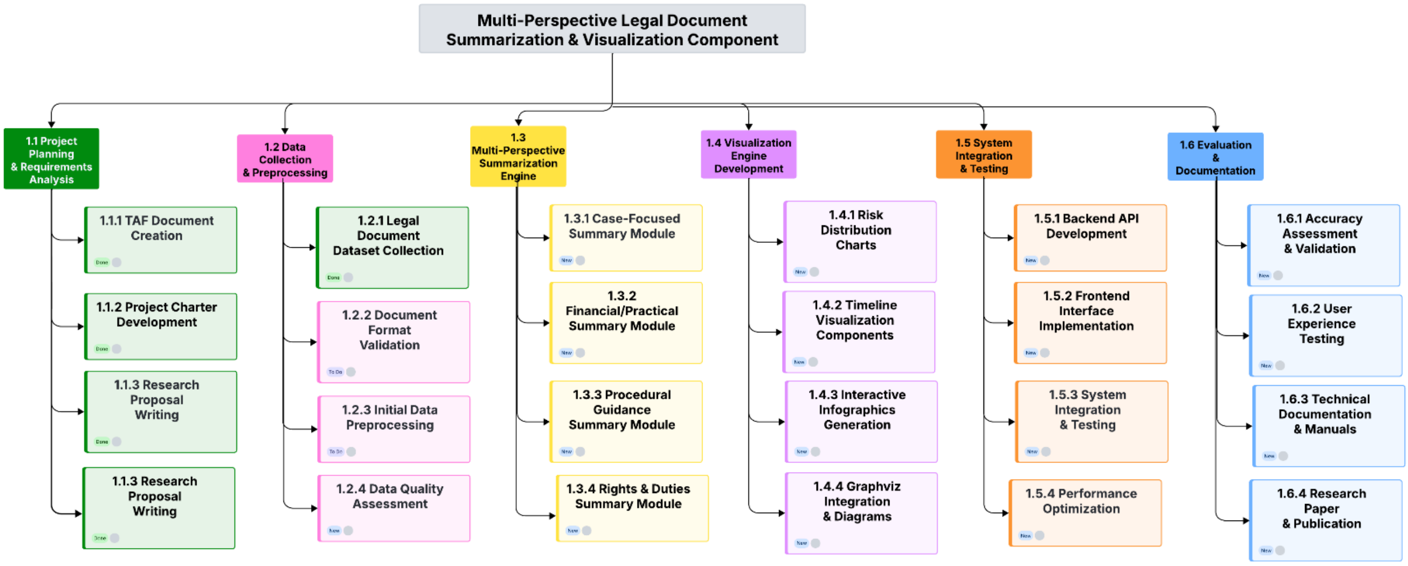

Figure 9.1: Work Breakdown Chart

The Figure 9.1: Work Breakdown Chart for the **Multi-Perspective Legal Document Summarization & Visualization Component** outlines the major project phases and tasks. The **green-colored tasks** (TAF Document Creation, Project Charter Development, and Research Proposal Writing) are already **completed**, marking the initial project planning and requirement analysis as finished. All other tasks across data collection, preprocessing, summarization engine development, visualization, system integration, and evaluation remain in the **to-do list**.

10.BUDGET AND BUDGET JUSTIFICATION

Component	Cost (LKR)	Justification
Travel & Meetings with Law College Experts	10,000	To consult legal experts for guidance and acquire datasets. Covers transport, refreshments, and coordination.
Data Collection	5,000	Purchase or access rights for legal datasets required for analysis and model training.
Research Paper Publication	100,000	Covers fees for publishing in reputed journals/conferences.
Internet Charges.	2,000 × 12 = 24,000	For continuous internet access required for dataset downloads, literature surveys, virtual meetings, and cloud-based experiments.

Table 1:Budget Justification

Total Estimated Budget: 139,000 LKR

The proposed budget covers key needs for the research. Travel and meeting costs support consultations with legal experts and dataset collection. Data collection ensures access to reliable legal texts for training and analysis. Publication costs are essential for sharing findings in reputed journals. Internet charges guarantee uninterrupted research, collaboration, and cloud-based experiments. Overall, the budget is balanced and focused on both research execution and dissemination.

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